

The newfloat package^{*}

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Abstract

This package offers the command `\DeclareFloatingEnvironment` for defining new floating environments which behave like `figure` and `table`.

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^{*}This package has version number v1.0, last revised 2011/10/30.

1 Loading the package

First of all you need to include this package into your document with

```
\usepackage[<options>]{newfloat}
```

where *<options>* are one or more of

```
within=<"within" counter> or none  
chapterlistsgap=<value>
```

The “within” counter specifies the counter which will be used to reset the counter of the floating environments `figure` and `table`. (Furthermore this setting will be used as default setting for `\DeclareFloatingEnvironment`.)

So for example `within=chapter` will give you a numbering scheme *<chapter>.<x>* for floating environments, while `within=section` will give you a numbering scheme *<chapter>.<section>.<x>*, or *<section>.<x>* if the document class does not offer `\chapter`. `within=none` will result in a continuous numbering throughout the document, i.e. the numbering scheme will be simply *<x>*.

The option `chapterlistsgap=<value>` sets the amount of the vertical gap inserted into the “List of Figure”, “List of Tables”, and all lists created with `\DeclareFloatingEnvironment` when a new chapter will be started. The default value is 10pt. (This option will only be available if the document class used offer the usage of chapters, e.g. the book or report document class.)

Both options can be changed later on, too, by using the command

```
\newfloatsetup<options> ,
```

2 \DeclareFloatingEnvironment

After loading the `newfloat` package you can define your own floating environments with

```
\DeclareFloatingEnvironment[<options>]{<type>}
```

where *<options>* are one or more of

```
fileext=<file extension>  
listname=<list name>  
name=<prosa name>  
placement=<combination of htbtp>  
within=<"within" counter> or none  
chapterlistsgaps=on or off
```

If no *<options>* are given, “lo*<type>*” will be used as *<file extension>* for the list, “List of *<name>*s” as *<list name>*. “*<name>*” as *<name>* (but with the first letter capitalized), “`tbp`” as *<placement>* specifier, and “chapter” resp. “none” as *<"within" counter>*, i.e., the counter which resets the numbering.

The default value of the `chapterlistsgaps=` option depends on the “within” setting, it is set to `on` if `chapter` or `section` is selected, otherwise it is set to `off`. (This option will only be available if the document class used offer the usage of chapters, e.g. the `book` or `report` document class.)

The list will be typeset using the command `\listof<type>s` resp. `\listof<type>es`, analogous to `\listoffigures` and `\listoftables`.

If the `fltpage` package is loaded, an environment called `FP<type>` will be defined additionally, same for `sideways<type>` (rotating package), `SC<type>` (sidecap package), and `wrap<type>` (wrapfig package).

So for example

```
\DeclareFloatingEnvironment{diagram}
```

will define a new floating environment called `diagram`, the list will be stored in a file with the extension `lodiagram`, the name (used for the caption) will be “Diagram” and the list name “List of Diagrams”. The list could be typeset with `\listofdiagrams`. Dependent on which packages are loaded, the environments `FPdiagram`, `sidewaysdiagram`, `SCdiagram`, and `wrapdiagram` will be defined additionally.

Another example:

```
\DeclareFloatingEnvironment[
  fileext=lox,
  listname={List of Matrixes},
  name=Matrix,
  placement=p,
  within=section,
  chapterlistsgaps=off,
]{matrix}
```

will define a new floating environment called `matrix` with the given settings. Please note that names which contain spaces needs to be enclosed in curly braces.

3 \SetupFloatingEnvironment

While `\DeclareFloatingEnvironment` will create new floating environments,

```
\SetupFloatingEnvironment{<floating environment>}{<options>}
```

will change the settings of existing ones, i.e. either `figure` or `table`, or a one created with `\DeclareFloatingEnvironment`, or a one created with `\newfloat` offered by the `float` package, or a one created with `\newfloat` offered by the `memoir` document class, or a one created with `\DeclareNewFloatType` offered by the `floatrow` package, or ...

The `<options>` are the same as the options for `\DeclareFloatingEnvironment`, but one should avoid changing the file extension of existing floating environments, i.e. using the `fileext=` option within `\SetupFloatingEnvironment` is usually a very bad idea.

An example:

```
\SetupFloatingEnvironment{lstlisting}{chapterlistsgaps=off}
```

will switch off the chapter lists gaps for `lstlisting` environments offered by the `listings` package.

4 \ForEachFloatingEnvironment

```
\ForEachFloatingEnvironment<code with #1>
```

will execute the given *<code>* for all known floating environments, and for ones defined with `\DeclareFloatingEnvironment` later on.

So for example the subcaption packages uses

```
\ForEachFloatingEnvironment{\DeclareCaptionSubType{#1}}
```

for initializing itself for all floating environments which are known to the `newfloat` package.

There is also a starred variant `\ForEachFloatingEnvironment*` which will execute the given code for already existing floating environments only, i.e. no hook will be placed inside `\DeclareFloatingEnvironment`.

An example:

```
\ForEachFloatingEnvironment*{\typeout{#1}}
```

will `\typeout` the names of all already known floating environments to the terminal and log file.

References

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